

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

```

```

if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
for i = 2:numel(fn)
    selectCuts = fn{i};
    if contains(selectCuts, 'msk')
        continue
    end
    if contains(selectCuts, 'all_fit_par')
        continue
    end

    fprintf('Processing cuts: %s \n', selectCuts);
    %selectCuts= 'Ei400_off200GPmaxFF';
    cuts_list = GP_FFCutsFits.(selectCuts);
    if ~iscell(cuts_list)
        cuts_list = {cuts_list};
    end
    if contains(selectCuts, 'Ei200')
        mi = maps_instrument(200,600, 'S');
    end
    if contains(selectCuts, 'Ei400')
        mi = maps_instrument(401,600, 'S');
    end
    if contains(selectCuts, 'Ei800')
        mi = maps_instrument(786,600, 'S');
    end
    sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
    for j=1:numel(cuts_list)
        cuts_list{j} = cuts_list{j}.set_instrument(mi);
        cuts_list{j} = cuts_list{j}.set_sample(sample);
        plot(cuts_list{j}); lz 0 1; keep_figure;
    end
    dir_name = "GP";

    dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
    half_dE = 5; % half width of data binning

    en_mm = cuts_list{1}.data.img_range(:,4);

```

```

en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
cl = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

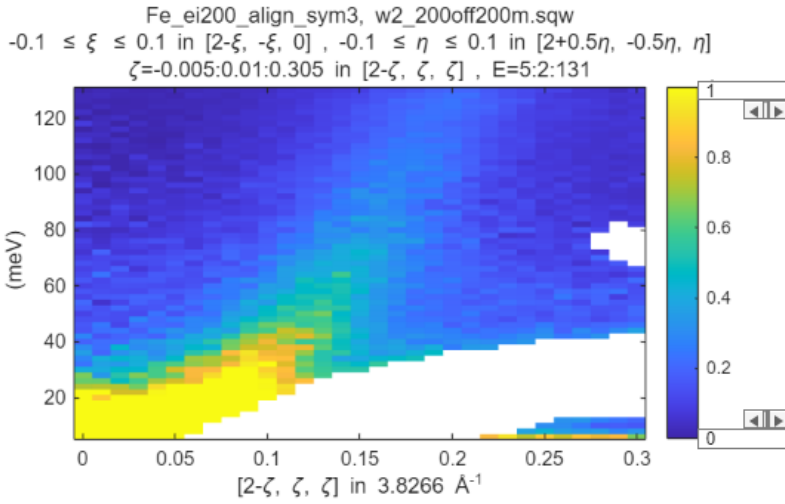
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure;

%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fig
ure

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
close all;
end

```

Processing cuts: Ei200_off200Gpmiff



**** En = 35±5 going to 125

**** En = 45±5 going to 125

**** En = 55±5 going to 125

**** En = 65±5 going to 125

**** En = 75±5 going to 125

**** En = 85±5 going to 125

**** En = 95±5 going to 125

**** En = 105±5 going to 125

**** En = 115±5 going to 125

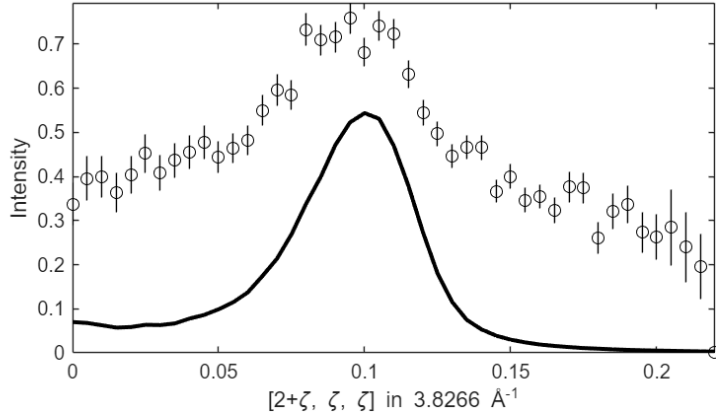
**** En = 125±5 going to 125

Processing cuts: Ei200_off200GPmaff

**** En = 35±5 going to 145

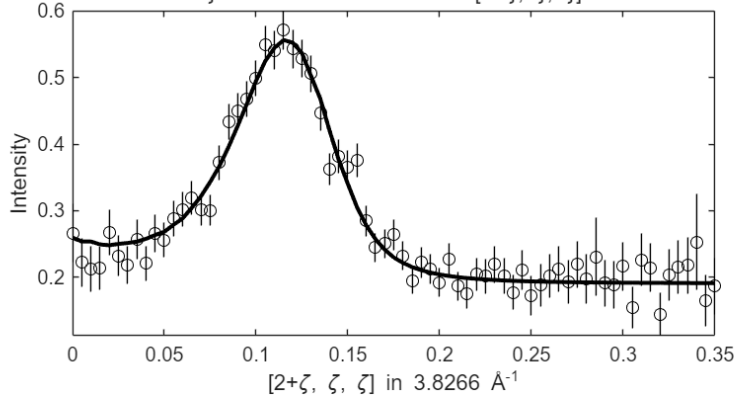
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.4085

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $30 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

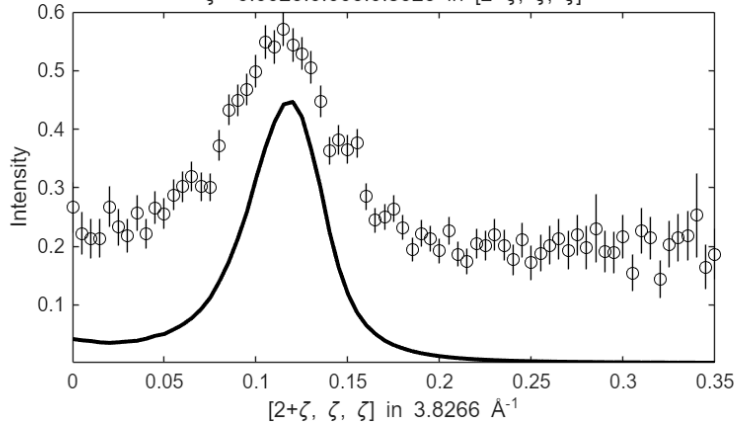


**** En = 45±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.97 \pm 0.054$; $J_0 = 24 \pm 0.24$; $\gamma = 9.00123 \pm 1.0409$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $40 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

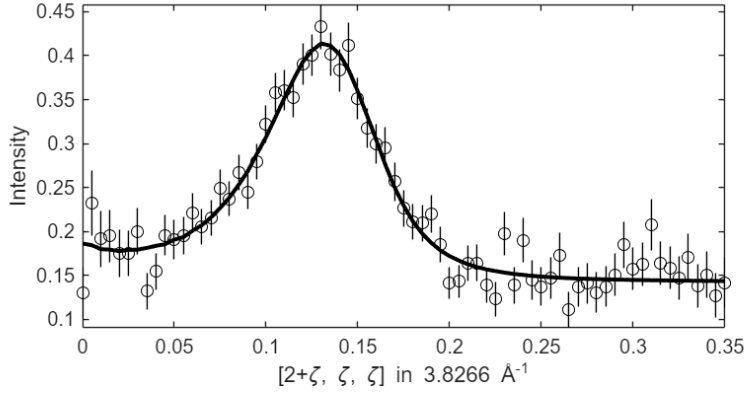


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $40 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

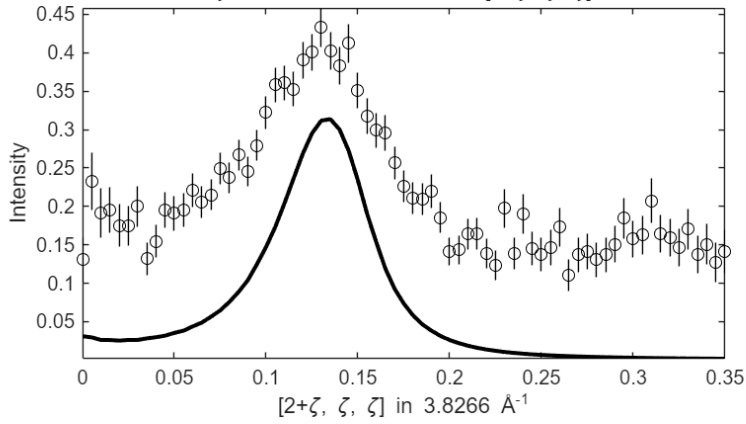


**** En = 55±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.97\pm0.026$; $J_0= 25\pm0.27$; $\gamma=14.3097\pm0.278775$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $50 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

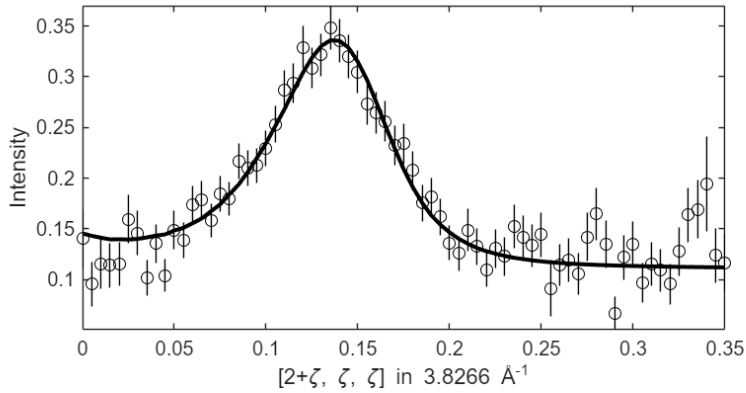


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.99\pm0.085$; $J_0= 28\pm0.42$; $\gamma=18.5005\pm2.38164$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

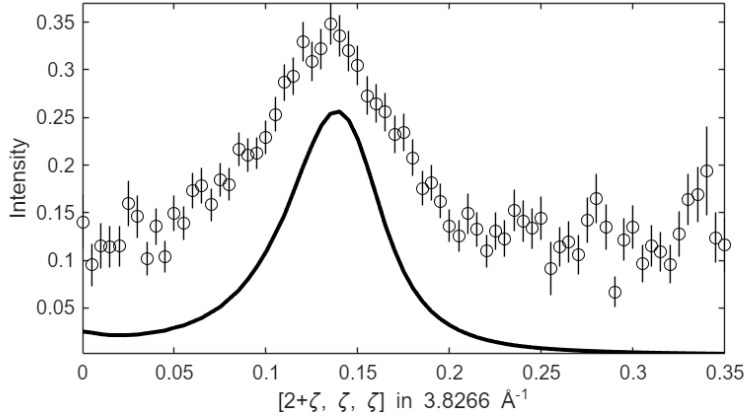


 **** En = 65±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.99\pm0.085$; $J_0= 28\pm0.42$; $\gamma=18.5005\pm2.38164$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

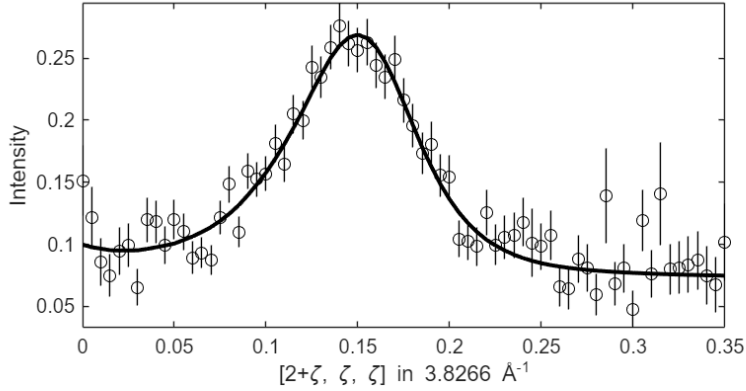


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

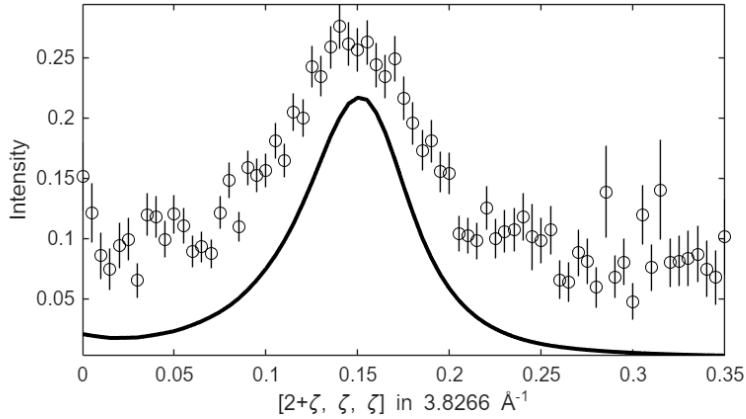


**** En = 75±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.1 \pm 0.094$; $J_0 = 28 \pm 0.41$; $\gamma = 23.7104 \pm 2.83847$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $70 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $70 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

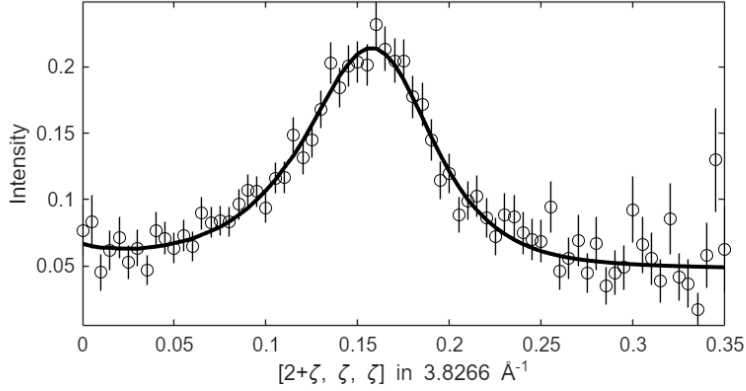


**** En = 85±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw

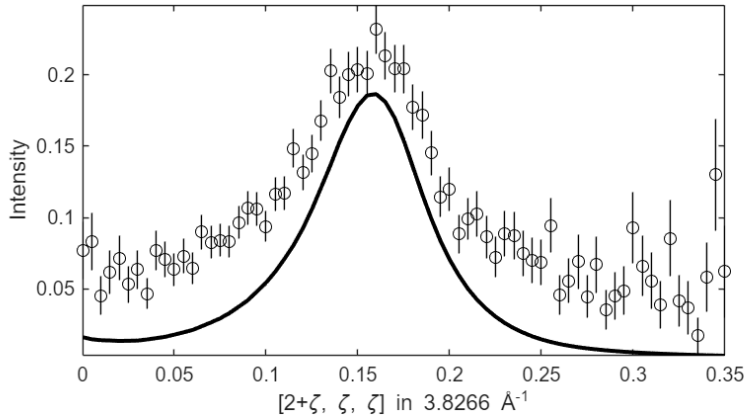
S= 1 ± 0.081 ; J0= 30 ± 0.37 ; gamma= 26.3464 ± 2.76335

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $80 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200ma.sqw

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $80 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

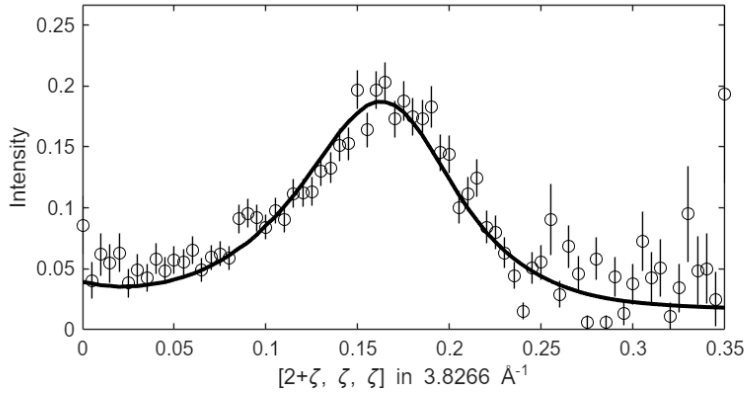


**** En = 95 ± 5 going to 145

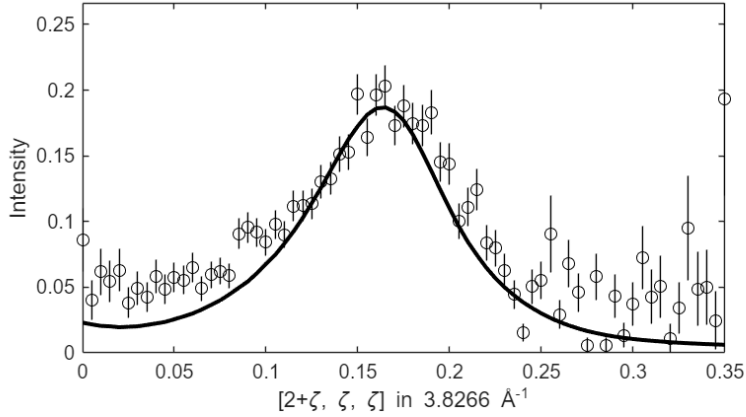
Fe_ei200_align_sym3, w2_200off200ma.sqw

S= 1.5 ± 0.2 ; J0= 32 ± 0.78 ; gamma= 39.8956 ± 6.51485

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $90 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

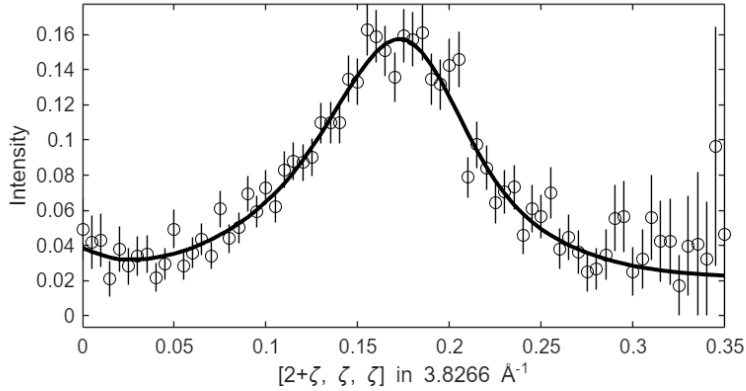


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $90 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

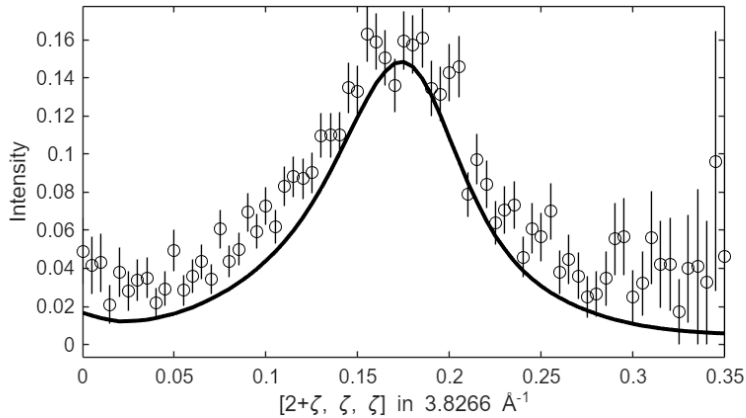


**** En = 105±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.3 \pm 0.11$; $J_0 = 33 \pm 0.46$; $\gamma = 40.1533 \pm 4.28932$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $100 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



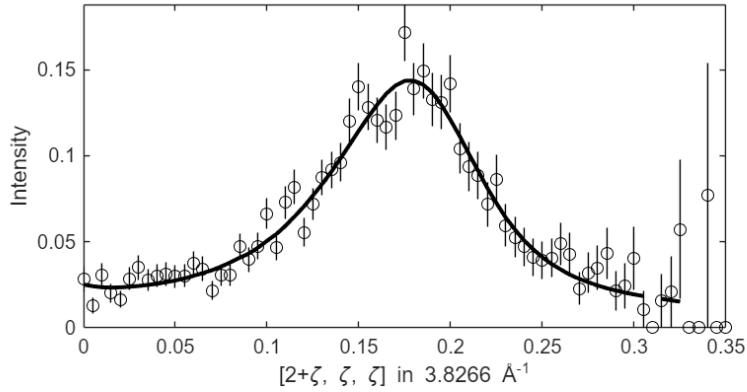
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $100 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



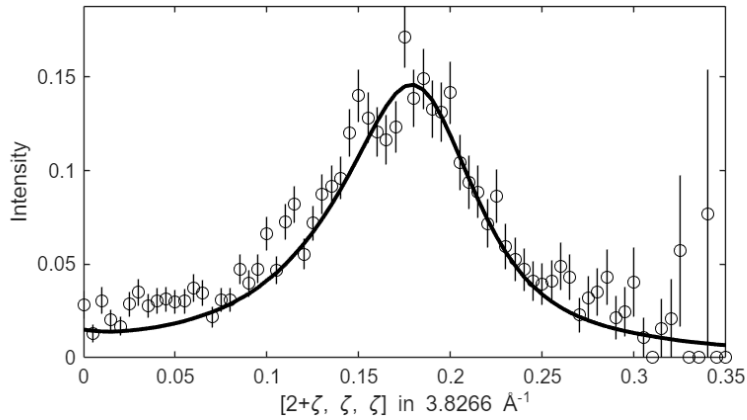
**** En = 115±5 going to 145

Warning: Dataset:5 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 7.0423

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.4 \pm 0.14$; $J_0 = 34 \pm 0.54$; $\gamma = 44.7125 \pm 5.22283$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $110 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



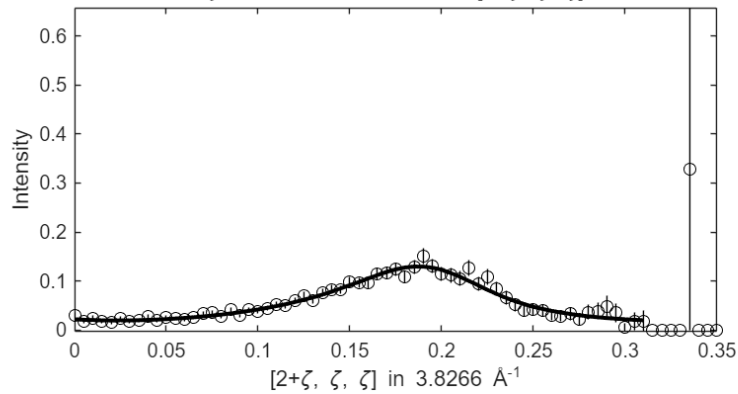
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.4 \pm 0.14$; $J_0 = 34 \pm 0.54$; $\gamma = 44.7125 \pm 5.22283$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $110 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



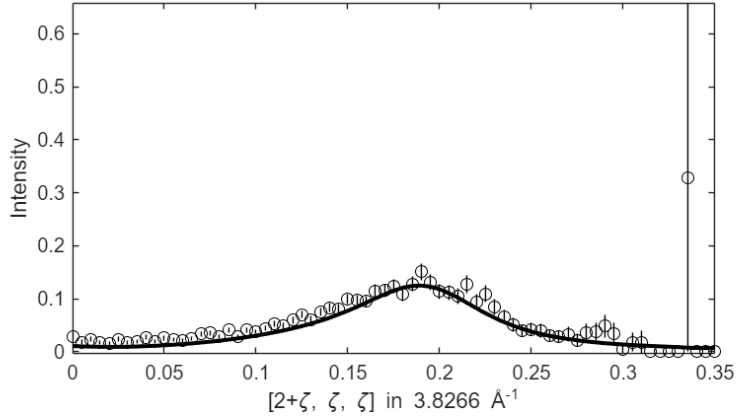
 **** En = 125±5 going to 145

Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 9.8592

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.3 \pm 0.065$; $J_0 = 34 \pm 0.41$; $\gamma = 45.9328 \pm 3.51629$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $120 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

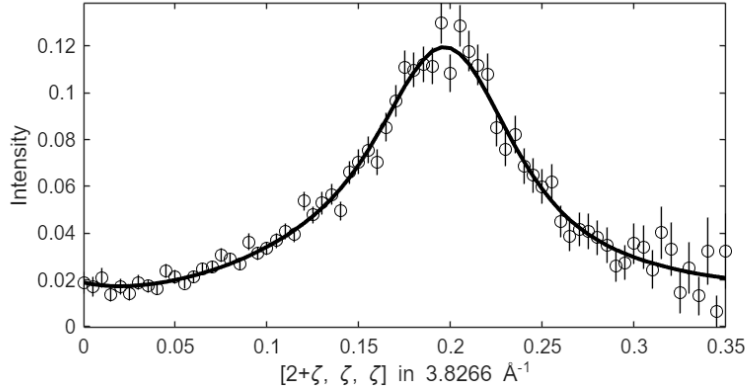


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $120 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$

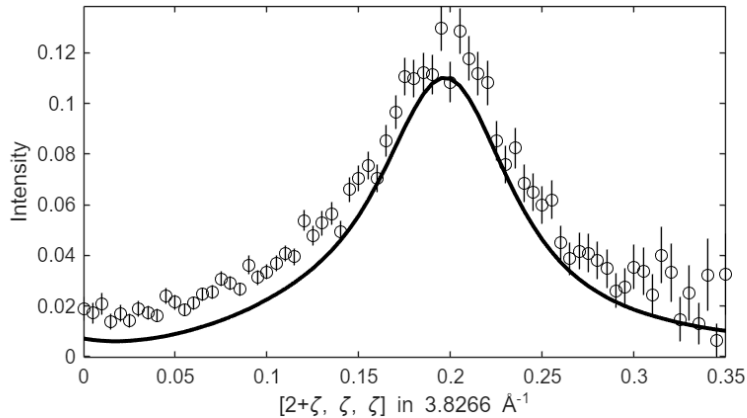


 **** En = 135±5 going to 145

Fe_ei200_align_sym3, w2_200off200ma.sqw
 S= 1.1±0.047; J0= 35±0.27; gamma=46.8211±2.56508
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $130 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



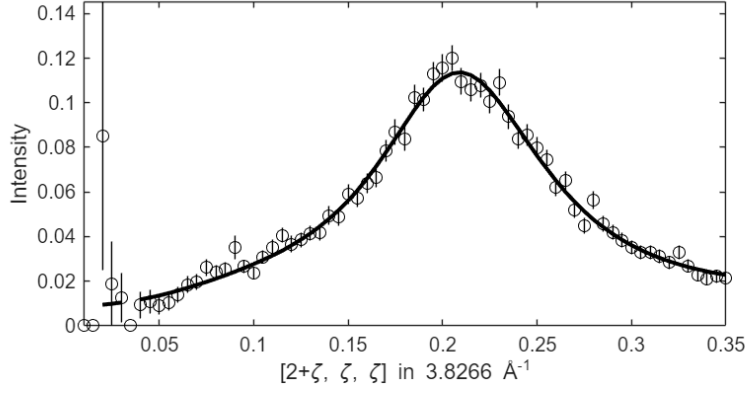
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $130 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



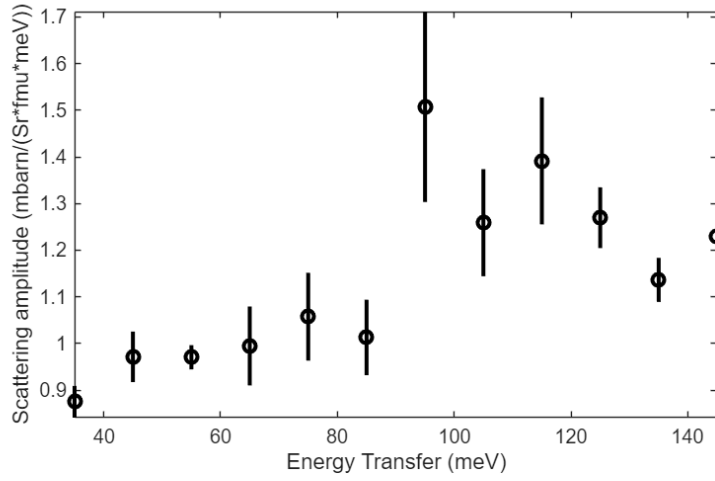
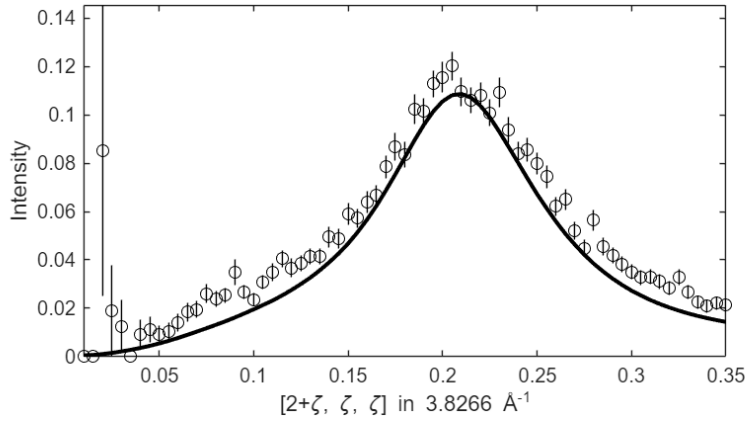
 **** En = 145±5 going to 145

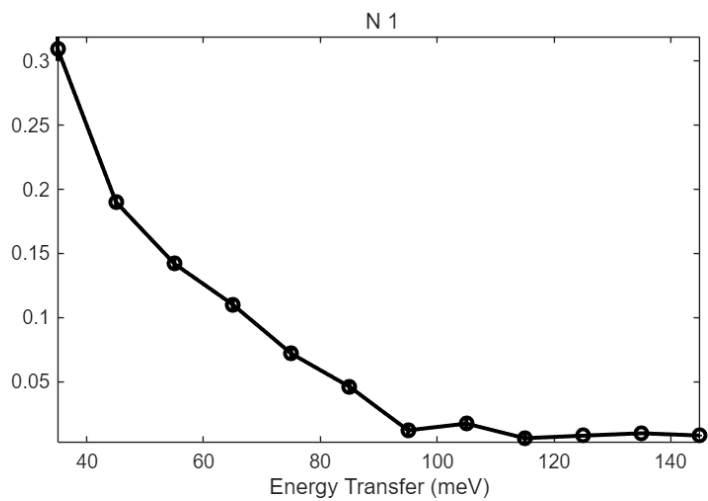
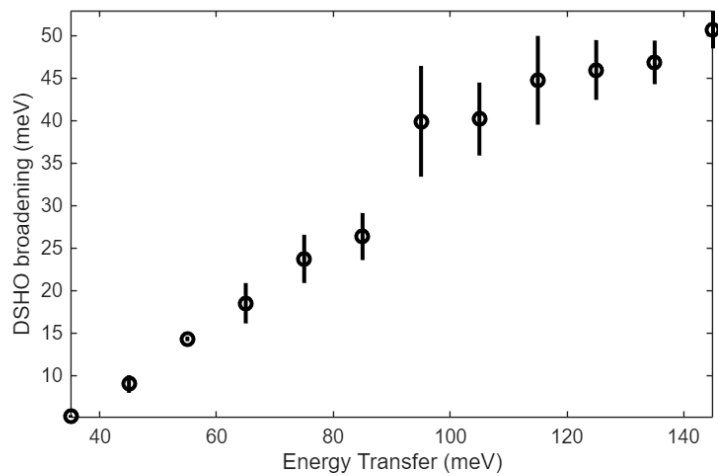
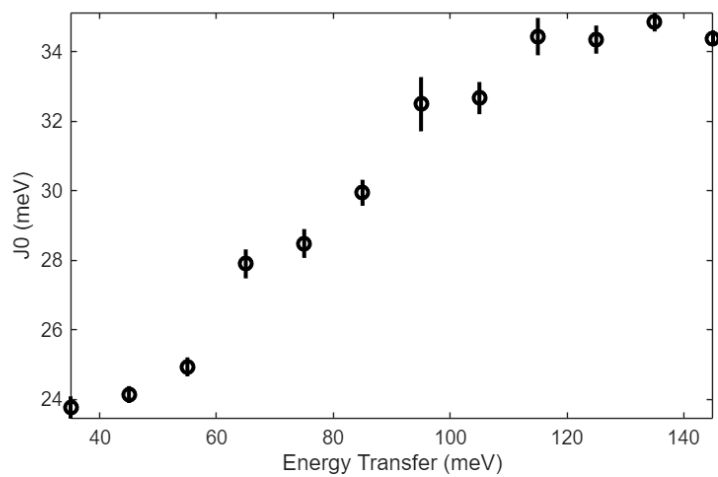
Warning: Dataset:3 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.2254

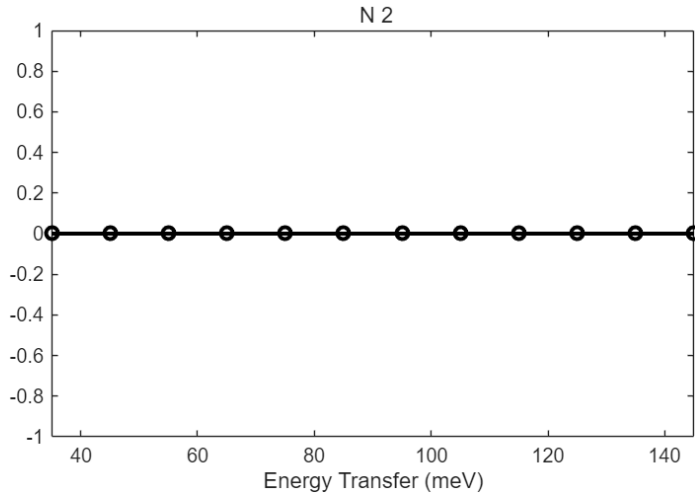
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.2 \pm 0.041$; $J_0 = 34 \pm 0.23$; $\gamma = 50.6902 \pm 2.21875$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $140 \leq E \leq 145$ meV
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200ma.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $140 \leq E \leq 145$ meV
 $\zeta = -0.0025:0.005:0.3525$ in $[2+\zeta, \zeta, \zeta]$





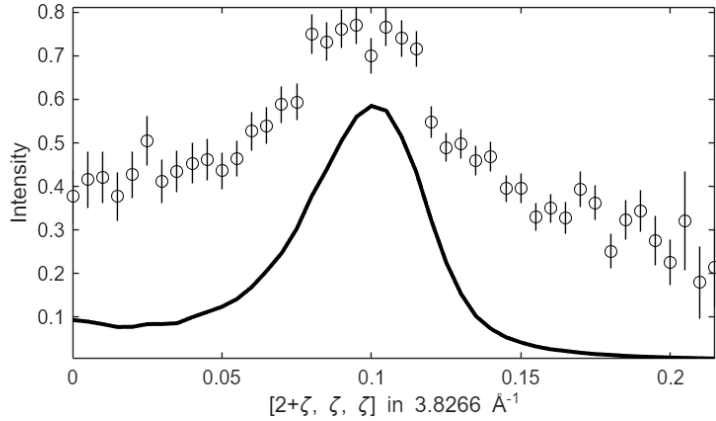


Processing cuts: Ei200_off110GPmaff

**** En = 35±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $30 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

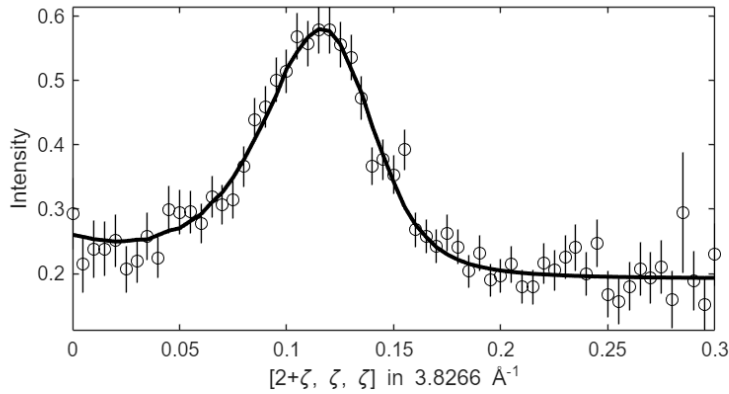


**** En = 45±5 going to 115

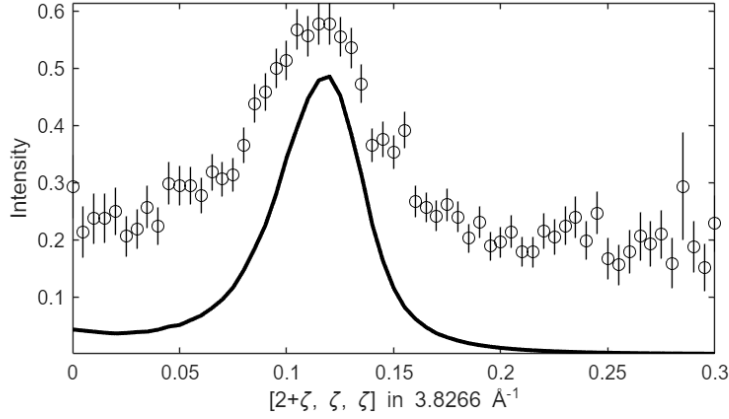
Fe_ei200_align_sym3, w2_200off200ma.sqw

$S=0.99 \pm 0.027$; $J_0 = 24 \pm 0.24$; $\gamma = 8.25974 \pm 0.176157$

$0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $40 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

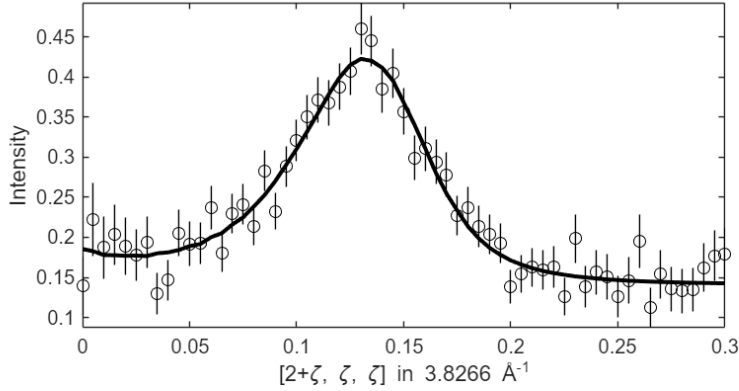


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $40 \leq E \leq 40$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

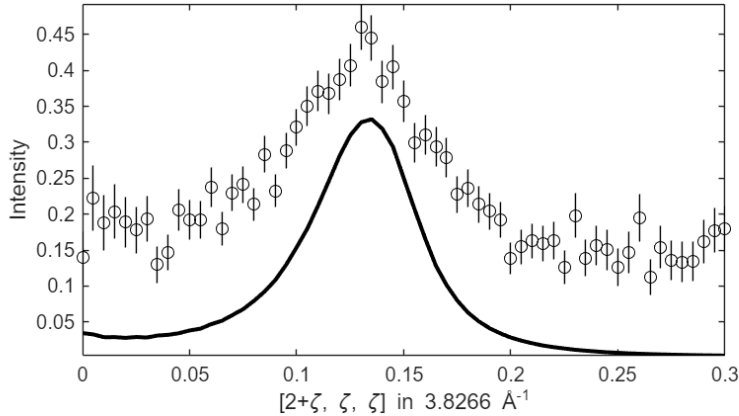


 **** En = 55±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1 \pm 0.032$; $J_0 = 25 \pm 0.3$; $\gamma = 14.1088 \pm 0.308514$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $50 \leq E \leq 50$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

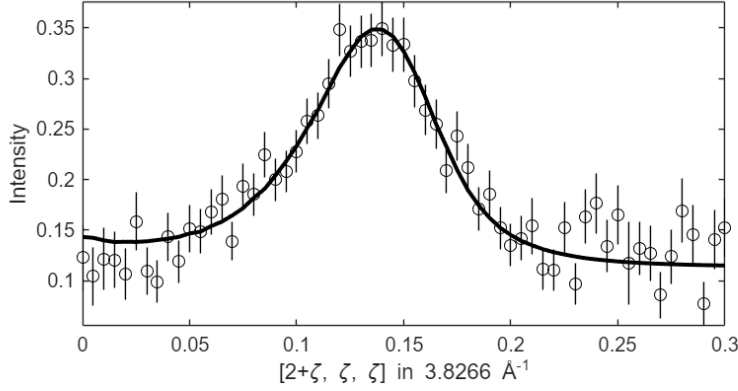


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $50 \leq E \leq 50$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

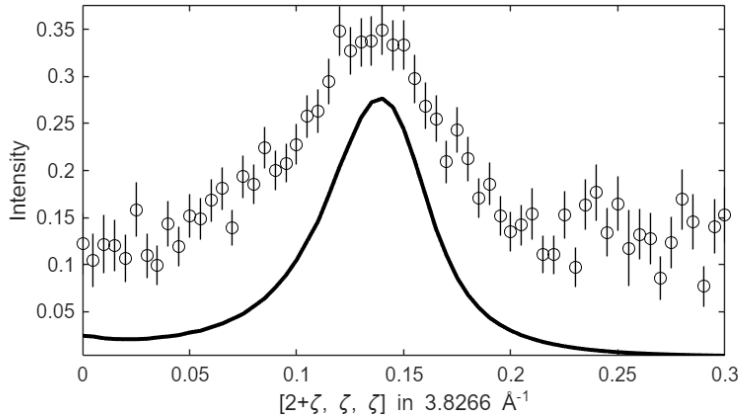


 **** En = 65±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S=0.96\pm0.098$; $J_0= 28\pm0.42$; $\gamma=16.127\pm2.54778$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

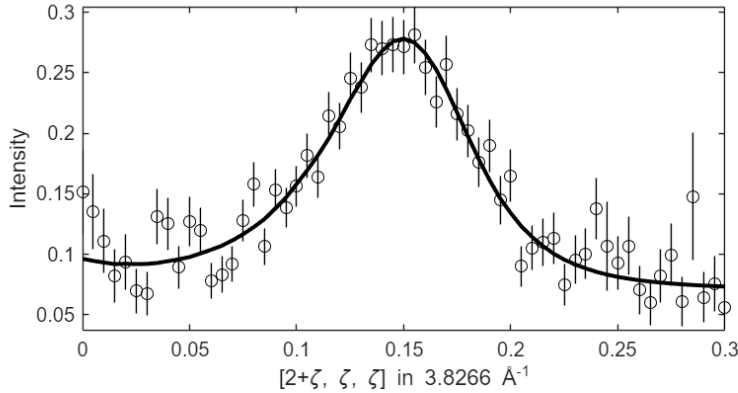


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $60 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

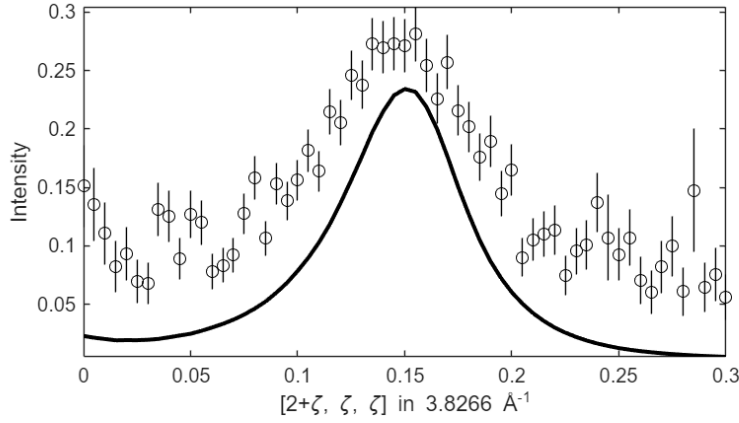


 **** En = 75±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S= 1.1\pm0.12$; $J_0= 28\pm0.47$; $\gamma=22.6574\pm3.4195$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $70 \leq E \leq$
 $\zeta=-0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

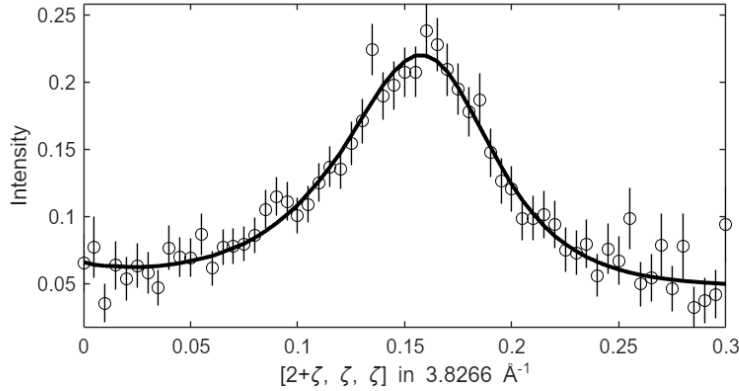


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $70 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

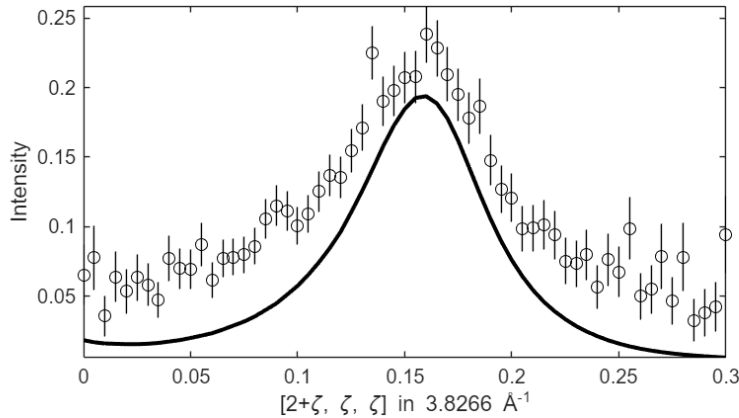


 **** En = 85±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.1 \pm 0.094$; $J_0 = 30 \pm 0.38$; $\gamma = 26.5462 \pm 2.96443$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $80 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$



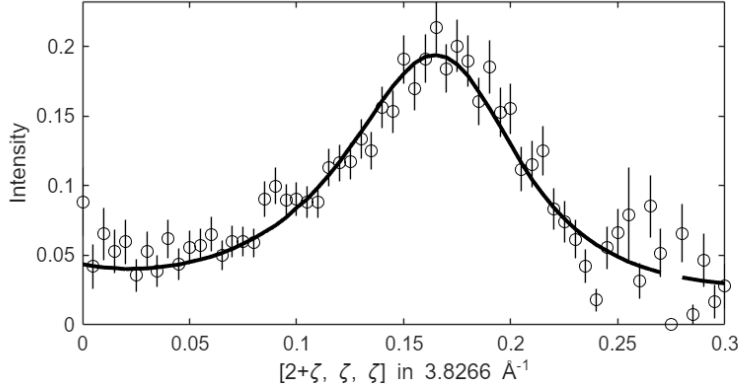
Fe_ei200_align_sym3, w2_200off200ma.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $80 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$



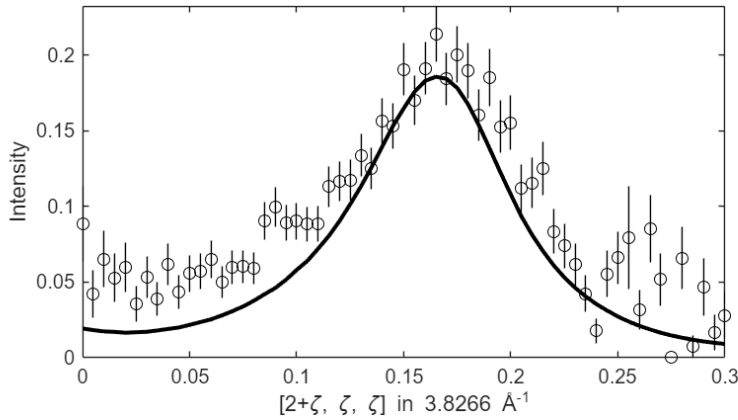
 **** En = 95±5 going to 115

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.6393

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.3 \pm 0.18$; $J_0 = 32 \pm 0.64$; $\gamma = 34.6689 \pm 5.53864$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $90 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

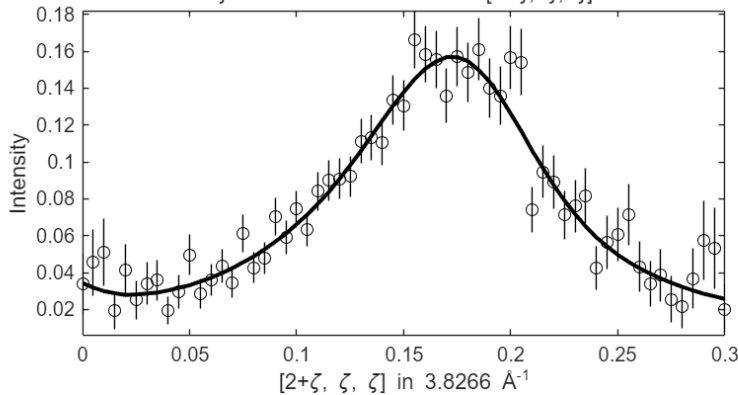


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.3 \pm 0.18$; $J_0 = 32 \pm 0.64$; $\gamma = 34.6689 \pm 5.53864$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $90 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

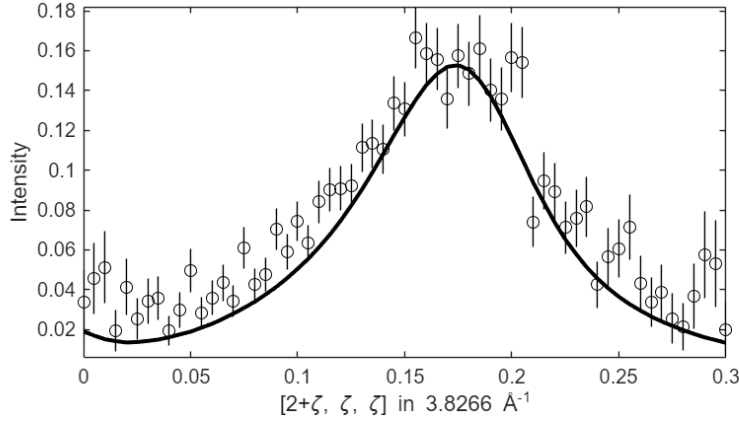


**** En = 105±5 going to 115

Fe_ei200_align_sym3, w2_200off200ma.sqw
 $S = 1.4 \pm 0.16$; $J_0 = 33 \pm 0.61$; $\gamma = 45.3673 \pm 5.74791$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $100 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

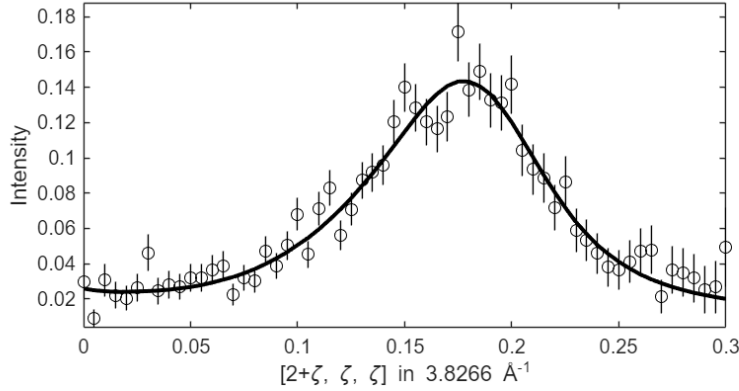


Fe_ei200_align_sym3, w2_200off200ma.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $100 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$

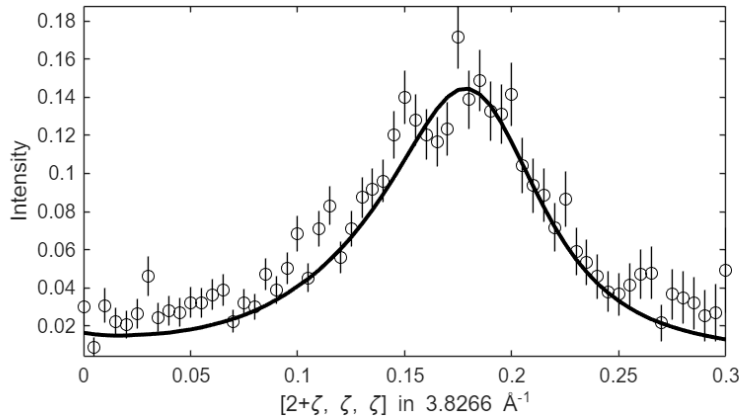


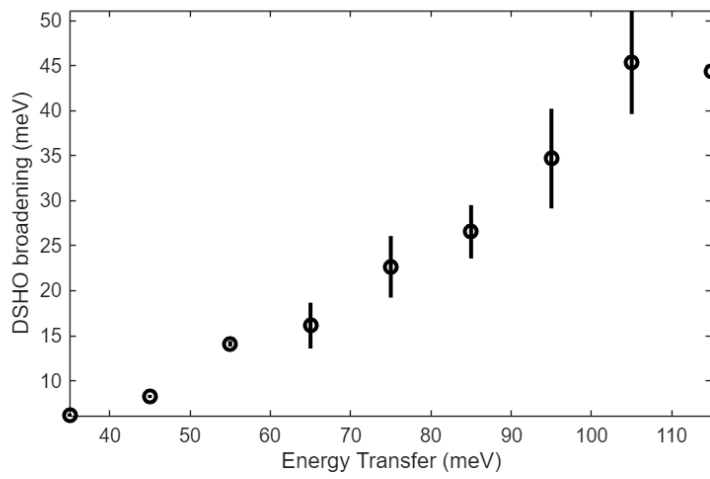
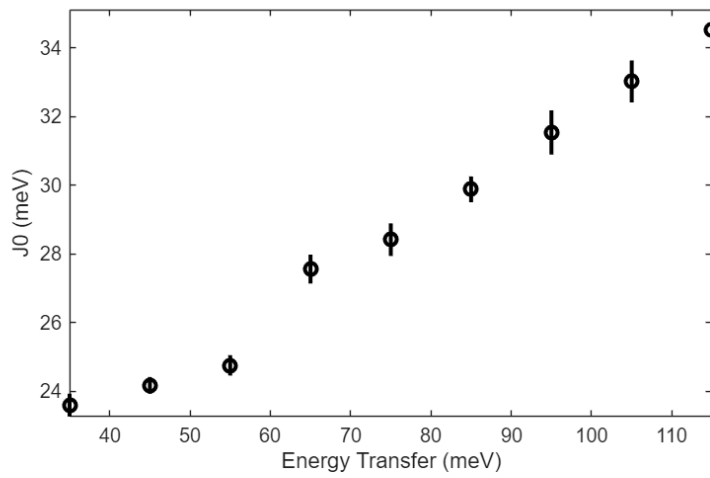
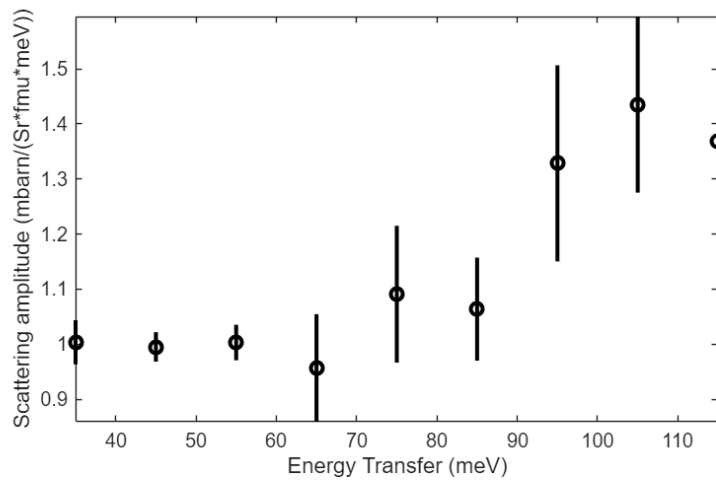
 **** En = 115±5 going to 115

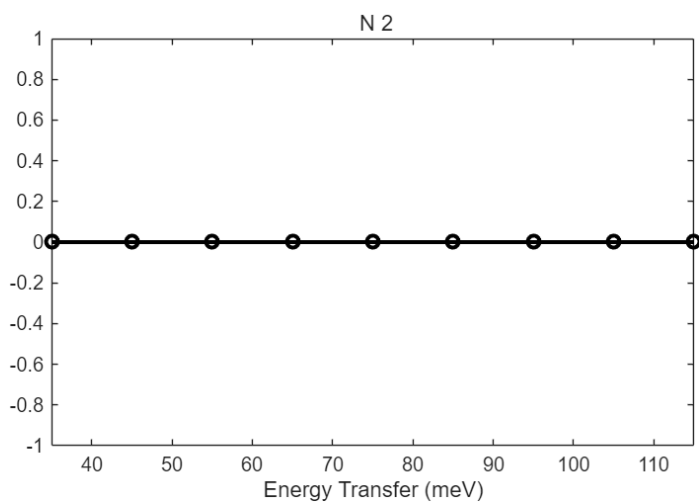
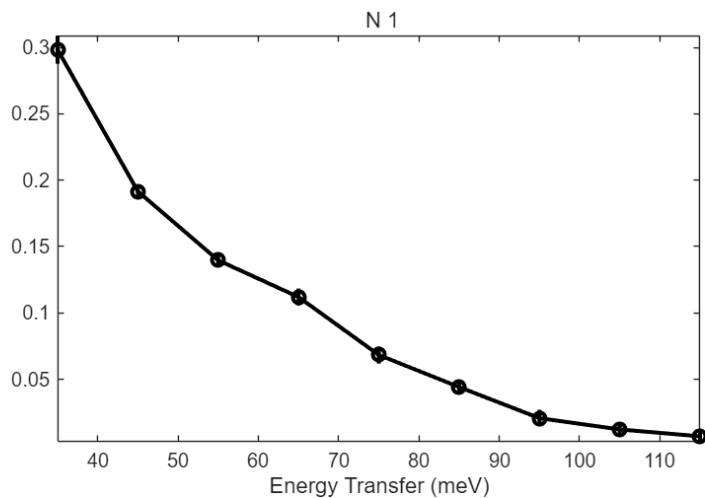
Fe_ei200_align_sym3, w2_200off200ma.sqw
 S= 1.4 ± 0.15 ; J0= 35 ± 0.58 ; gamma= 44.3942 ± 5.73226
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $110 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$



Fe_ei200_align_sym3, w2_200off200ma.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $110 \leq E \leq$
 $\zeta = -0.0025:0.005:0.3025$ in $[2+\zeta, \zeta, \zeta]$





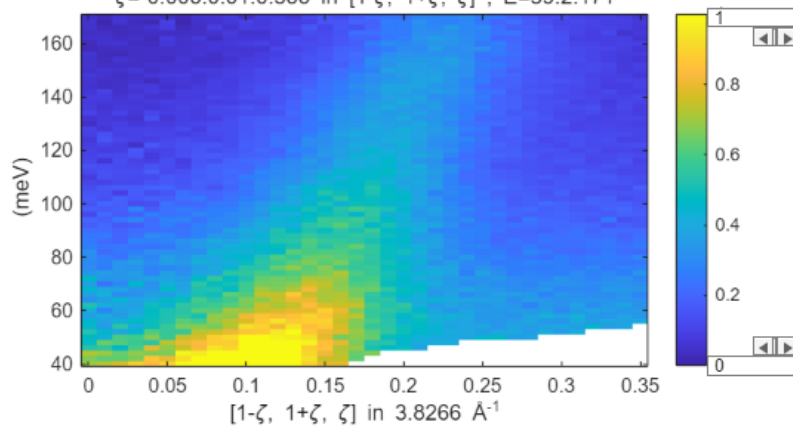


Processing cuts: Ei400_off110GPminFF

Fe_ei401_align_sym3, w2_400off110.sqw

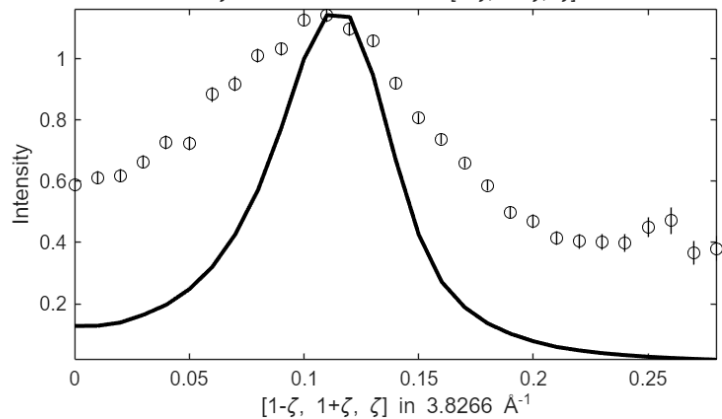
$-0.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$

$\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$, $E = 39:2:171$



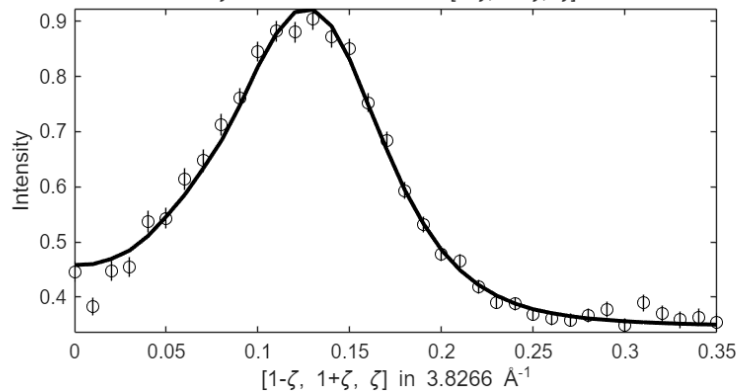
**** En = 44±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $39 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

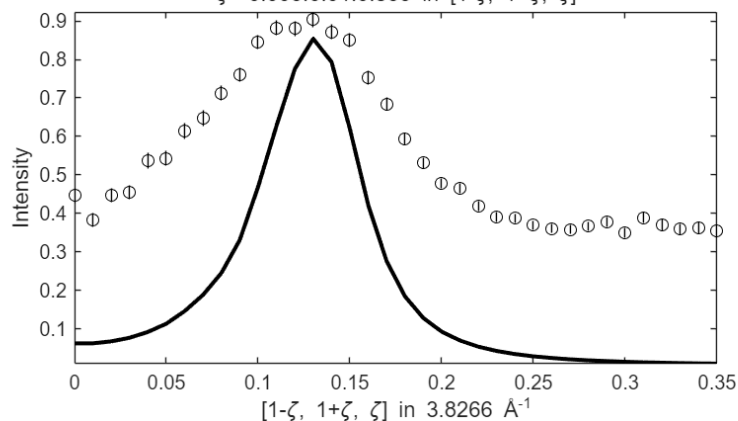


**** En = 54±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.91 \pm 0.015$; $J_0 = 25 \pm 0.23$; $\gamma = 16.7428 \pm 0.260224$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

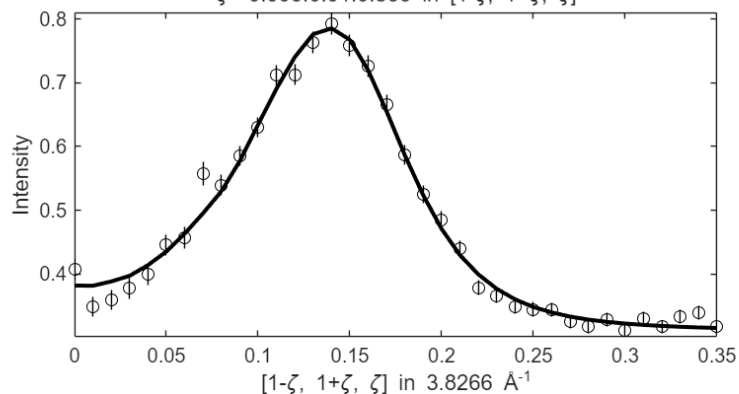


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

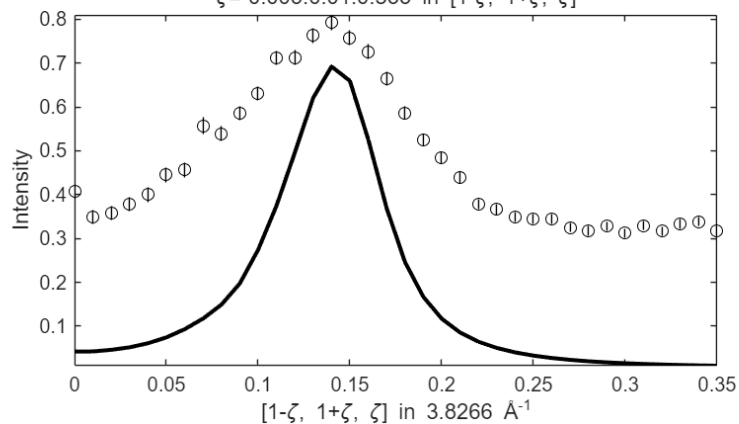


**** En = 64±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.82\pm0.013$; $J_0=27\pm0.23$; $\gamma=19.0564\pm0.272605$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

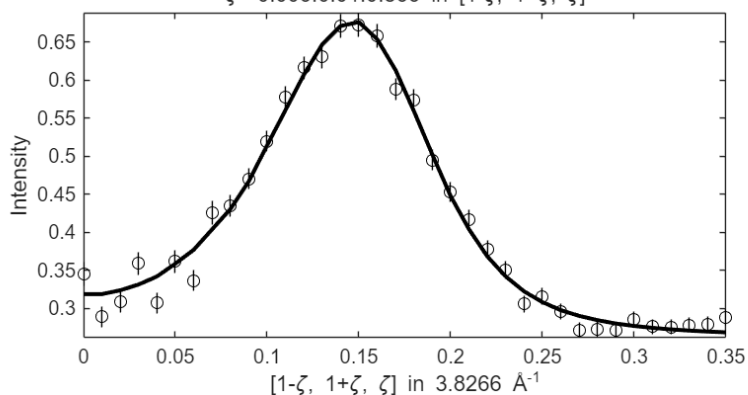


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

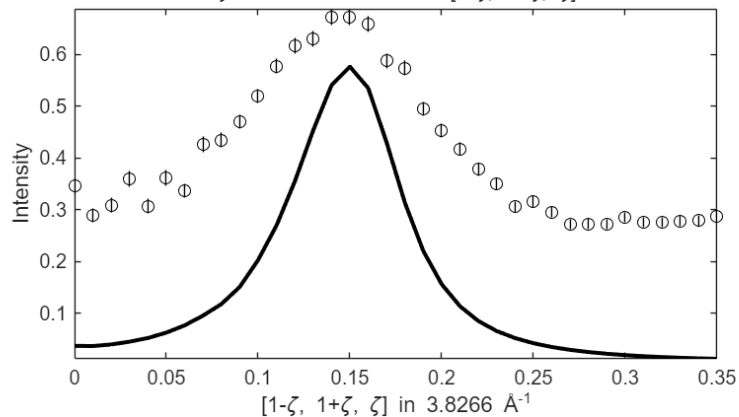


 **** En = 74±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.83\pm0.052$; $J_0=29\pm0.38$; $\gamma=23.6439\pm2.52872$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

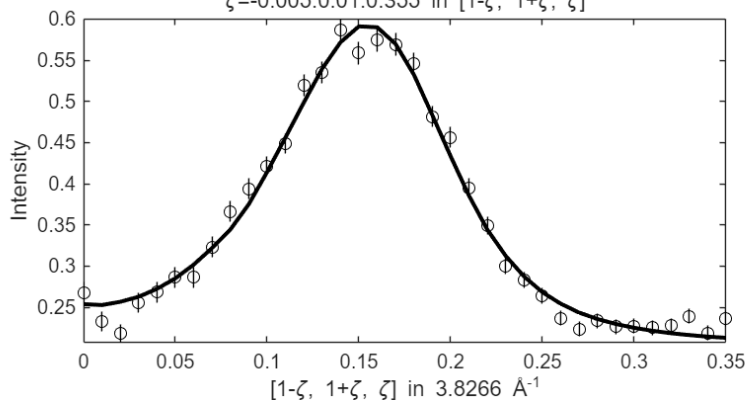


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

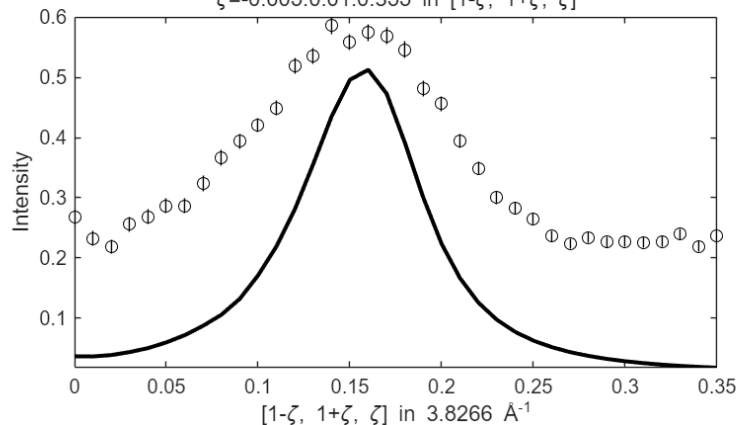


**** En = 84±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92 \pm 0.019$; $J_0 = 30 \pm 0.25$; $\gamma = 30.5106 \pm 0.437069$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

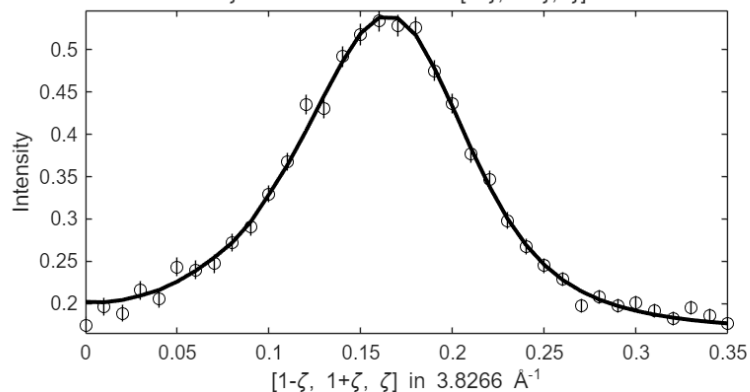


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

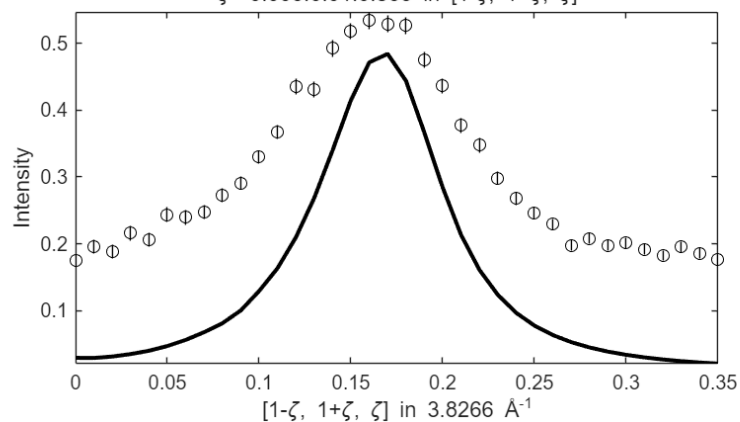


**** En = 94±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92\pm0.015$; $J0= 31\pm0.17$; $\gamma=32.2556\pm0.35767$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

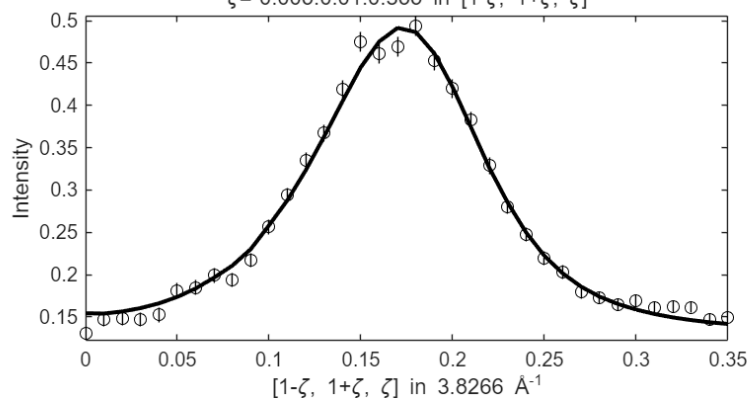


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

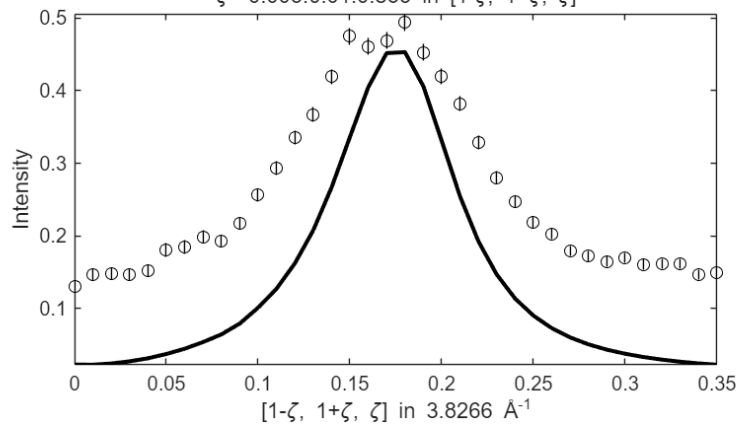


 **** En = 104±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.91\pm0.051$; $J0= 32\pm0.31$; $\gamma=33.7478\pm2.69375$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

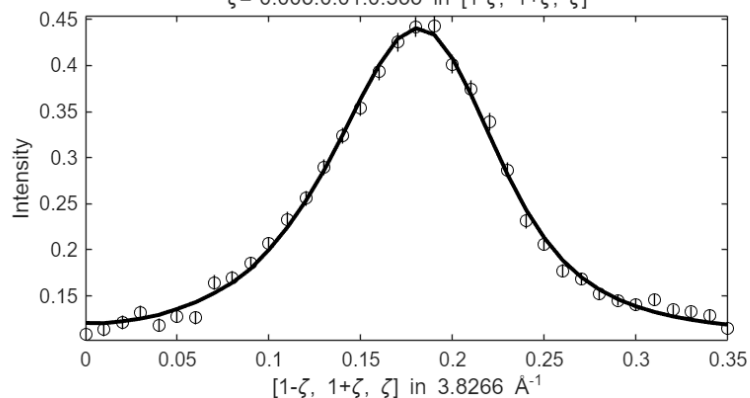


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

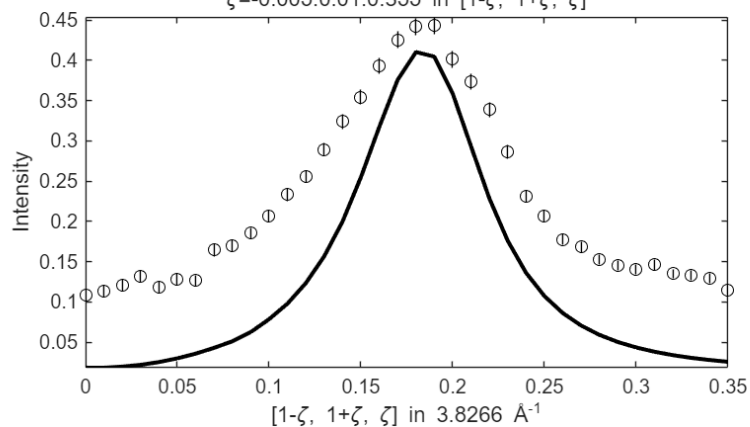


**** En = 114±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.89 \pm 0.018$; $J_0 = 32 \pm 0.14$; $\gamma = 36.4896 \pm 0.546682$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$:
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

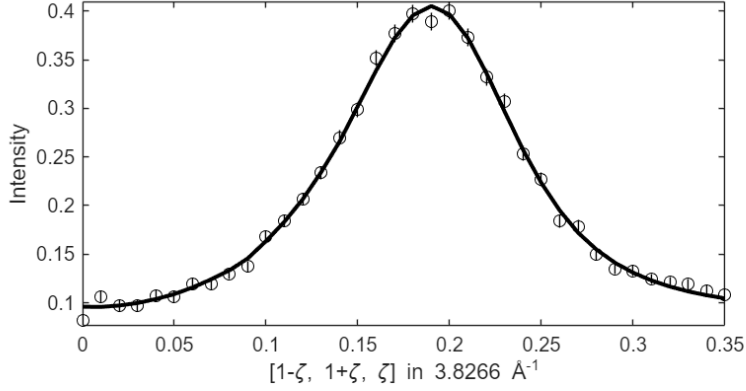


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E$:
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

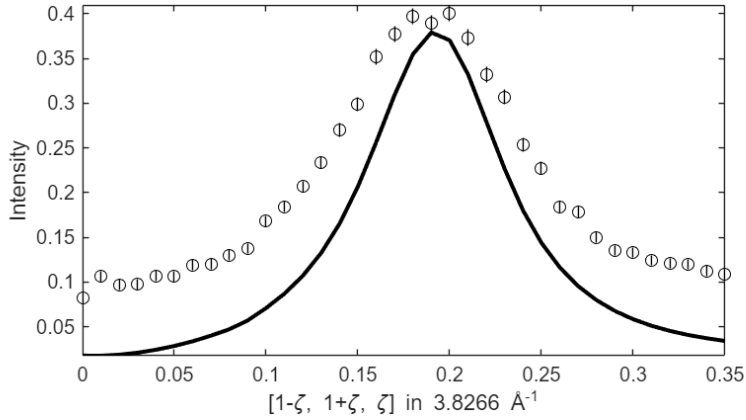


**** En = 124±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.95\pm0.019$; $J0= 33\pm0.12$; $\gamma=43.0297\pm0.70838$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

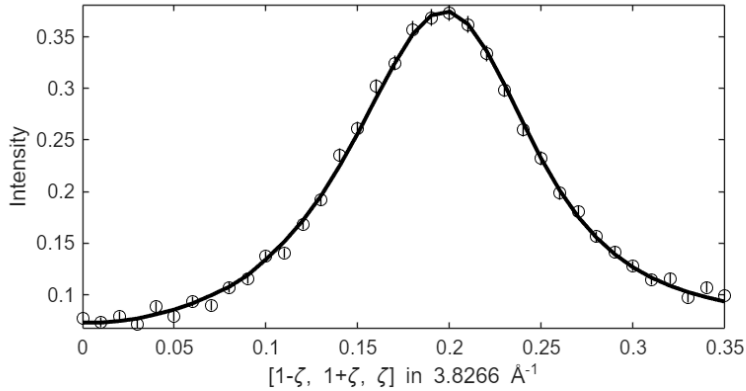


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

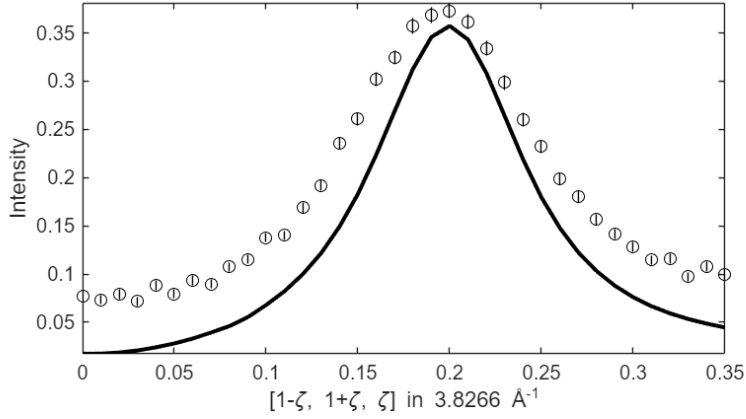


**** En = 134±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S= 1\pm0.028$; $J0= 35\pm0.14$; $\gamma=50.2021\pm1.35132$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

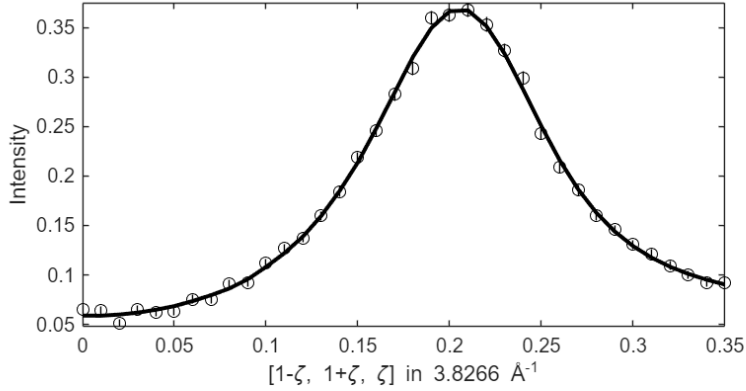


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

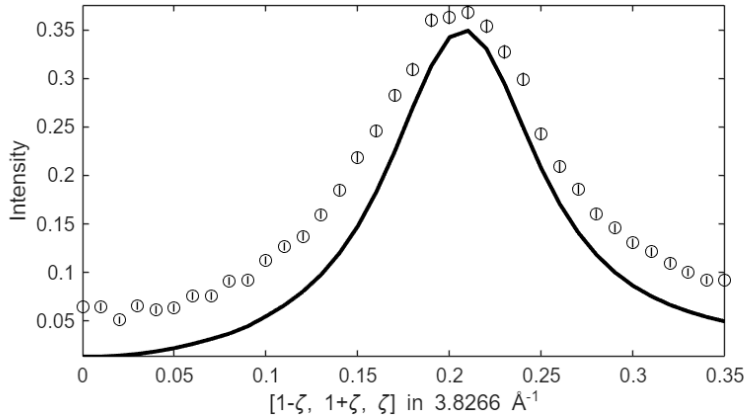


**** En = 144±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.023$; $J_0 = 35 \pm 0.13$; $\gamma = 49.6234 \pm 1.26538$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

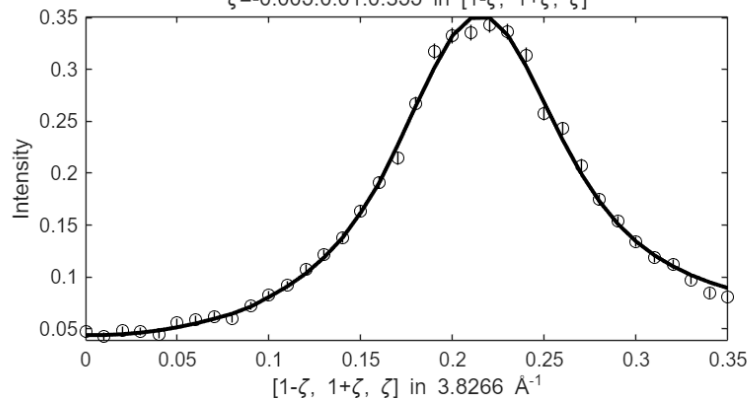


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

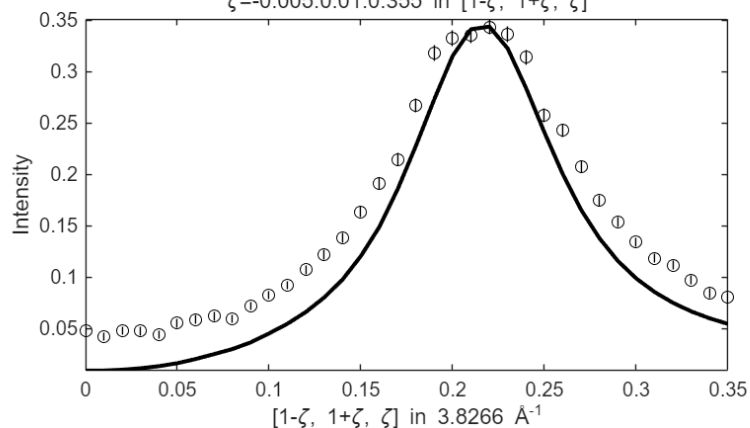


**** En = 154±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.026$; $J_0 = 35 \pm 0.16$; $\gamma = 50.2087 \pm 1.51809$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

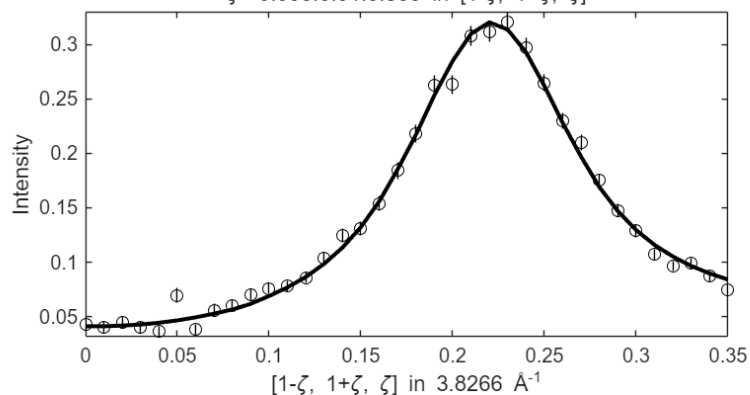


Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.026$; $J_0 = 35 \pm 0.16$; $\gamma = 50.2087 \pm 1.51809$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

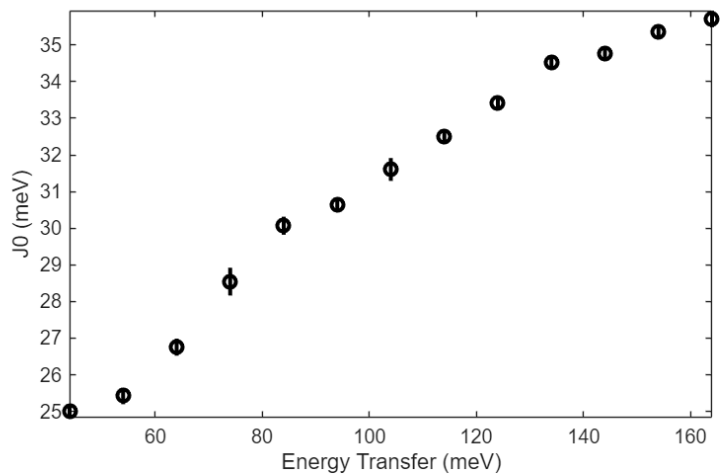
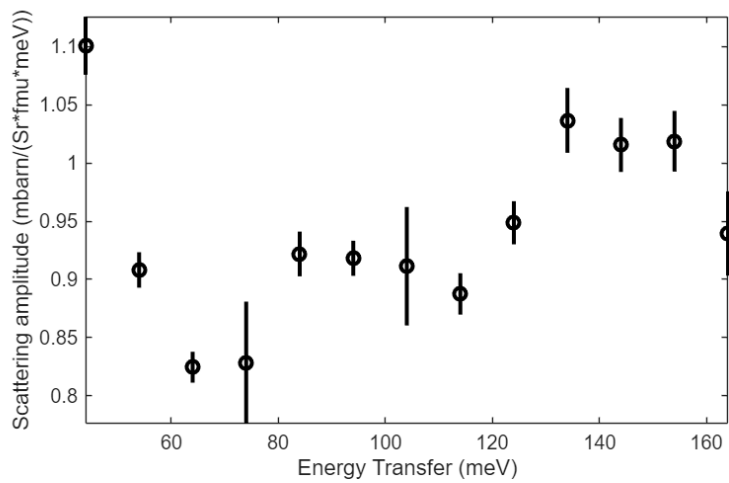
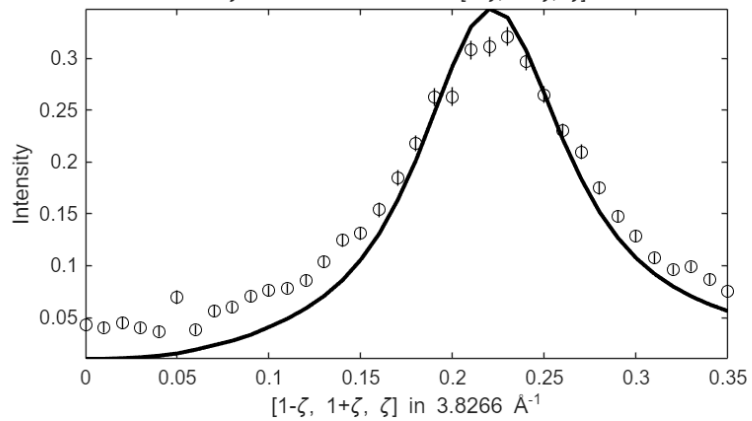


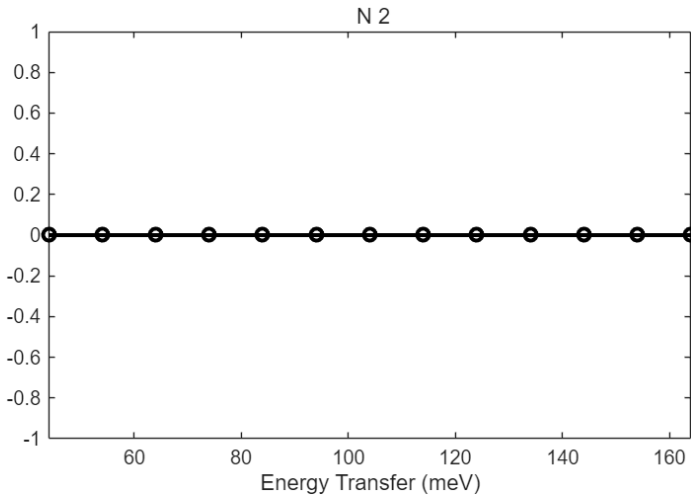
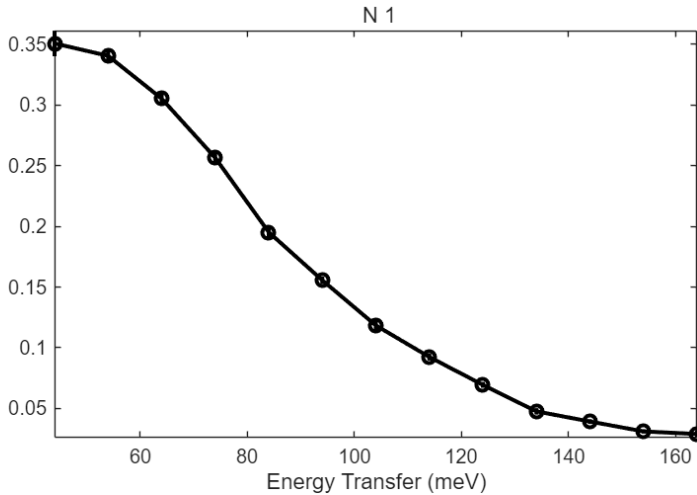
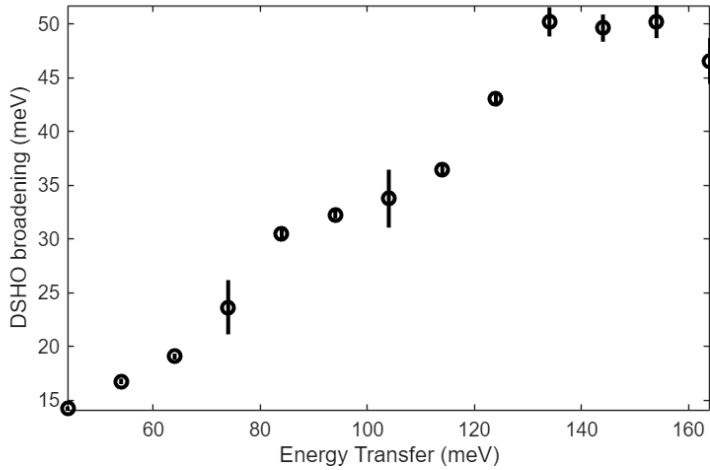
**** En = 164±5 going to 164

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 0.94 \pm 0.036$; $J_0 = 36 \pm 0.23$; $\gamma = 46.5719 \pm 2.13774$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off110.sqw
 $l \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$





```
Error using export>checkWritePermission (line 251)
Unable to overwrite existing file. Check write permission or whether the file is open in another application.
```

```
Error in export (line 104)
checkWritePermission(results.Destination);
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
fit_src_struct = GP_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```